Charge and the electric car

antocciphysics | TEACHER KEY | CPA

Learning target: Explain charge interactions and trace an electric car’s energy transfers.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 16; 3 course days later.*
It becomes 1/3 as large by Wien’s law.

02. Explain a worked example (worked)
Opposite changes -3 and +3; total charge conserved for the isolated pair.

03. Look and explain (model)
Like signs repel; opposite signs attract. Forces are equal in magnitude and opposite in direction.

04. Build the solution (practice)
Chemical → electrical → rotational → car kinetic energy; thermal energy and sound are possible outputs.

05. Use the idea (reason)
The same steady current passes each point; the motor transfers energy, not consuming charge.

06. Choose a method (mixed) *Review: Unit 6, Day 7; 12 course days later.*
+4 mm; partly cancel.

07. Exit check (exit)
Repulsion.

08. STRETCH: Apply it another way (stretch)
No; a charged object can attract a neutral polarizable object. Like net charges repel in the simple model.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 16; 3 course days later.*
T≈3000 K (about 3000 K); peak wavelength is rounded.

02. Explain a worked example (worked)
Opposite changes -3 and +3; total charge conserved for the isolated pair.

03. Represent and interpret (model)
Like signs repel; opposite signs attract. Forces are equal in magnitude and opposite in direction.

04. Solve and justify (practice)
90 J; charge circulates while energy is transferred.

05. Analyze the reasoning (reason)
The same steady current passes each point; the motor transfers energy, not consuming charge.

06. Choose a method (mixed) *Review: Unit 6, Day 7; 12 course days later.*
Resultant -10 mm. Energy also depends on motion, spatial gradients and the surrounding wave field.

07. Exit check (exit)
Polarization redistributes charge; nearer opposite charge can attract more strongly than farther like charge repels.

08. STRETCH: Apply it another way (stretch)
Shared positive charge gives +4,+4 nC; other remains -8 nC. Total zero.